

OMER ABDULAZIZ

Software Engineer

Harar, Ethiopia (EAT) | github.com/omer-biz | vesl.et

SUMMARY

Versatile polyglot engineer bridging the gap between high-level enterprise business logic and low-level systems architecture. Experienced in deploying complex operational software (Laravel/PHP) while specializing in memory-safe concurrency (Rust, C) and mathematically sound front-end development (Elm). Arctic Code Vault Contributor seeking opportunities to build robust infrastructure, developer tooling, or AI systems.

OPEN SOURCE & SYSTEMS PROJECTS

- ersha-os (Rust):** Building Digital Public Infrastructure (DPI) for the agricultural sector. Architected a highly concurrent, memory-safe backend designed for scale and high-impact civic utility in Ethiopia.
- amica (Rust / Lua):** Developed a high-performance proxy server using Rust for memory-safe networking, integrating Lua for a highly flexible, scriptable configuration layer.
- hermes (C):** Engineered a library of parser combinators purely in C, demonstrating a deep understanding of manual memory management, pointer arithmetic, and parsing theory.
- qeጒጒ (Elm):** Built and deployed an Amharic touch-typing web application (hosted at qelm.vesl.et). Utilized Elm's strict functional architecture to ensure a responsive, runtime-exception-free user interface.
- visu (Elm / Rust):** Created a developer tool to visually map and parse niri keymaps, optimizing Linux desktop environments.

PROFESSIONAL EXPERIENCE

UnifyTechnology

Full Stack Developer | Jan, 2025 – Present

- Modern SPA Architecture:** Architected and deployed comprehensive management systems (land/real estate, pharmacy, wholesale distribution) utilizing a **Laravel, React, and Inertia.js** stack.
- Seamless Data Flow:** Delivered high-performance Single-Page Applications (SPAs) by leveraging Inertia.js to bridge server-side Laravel routing with client-side React components, eliminating the need for complex REST API overhead.
- Database & System Logic:** Designed relational schemas and multi-tenant data structures to handle secure transactions and diverse user roles across mission-critical enterprise applications.

Harari Water Supply and Sewerage Authority

IT Infrastructure & Internal Tools Developer | Aug, 2024 – Jan, 2025

- Enterprise Communications:** Architected and deployed a bi-directional enterprise SMS messaging platform utilizing Python, FastAPI, and React to automate critical organizational communications.
- Internal Tooling:** Developed custom internal software to streamline workflows for utility employees, directly translating operational bottlenecks into automated technical solutions.
- Infrastructure Management:** Oversee and maintain the company's IT infrastructure, ensuring high availability, network security, and reliable deployments for mission-critical utility services.

SKILLS MATRIX

Systems & Backend: Rust, C, Lua, PHP, Laravel, Python

Frontend & UI: React, Inertia.js, Elm, TypeScript, JavaScript/HTML/CSS

Infrastructure & Tooling: Linux Administration, IT Operations, Distributed Systems, Parser Architecture, Git, PostgreSQL

Languages: English, Amharic, Arabic

EDUCATION

Addis Ababa Science and Technology University (AASTU)

Bachelor of Science in Software Engineering